PRACTICE EXAM 1 OF 4 · EXAM STRATEGY - ALL DOMAINS

Practice exam 1

Twenty questions across all four domains, weighted like the real exam.

QUESTIONS 20 TIME About 30 minutes MODULE 32 of 35

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How to take this exam

- **Answer before the reveal.** Say or write your letter while the question holds.
- **Keep a tally by question number,** so you can score by domain at the end.
- **Use the technique from module 31:** last sentence first, planted detail, eliminate.
- **Do not pause to look things up.** This is a measure, not a lesson.

> ON THE EXAM
>
> Your pattern of misses by domain matters more than your total.

> A new Pod has been created and stored, but its spec has no node assigned yet.

Which component will assign it to a node?

A The kubelet

B The kube-scheduler

C The kube-controller-manager

D kube-proxy

◆ PAUSE AND DECIDE

Which component will assign it to a node?

A The kubelet

Runs Pods already bound to its node.

B The kube-scheduler

Watches for unassigned Pods and binds each to a node.

C The kube-controller-manager

Creates Pods; does not place them.

D kube-proxy

Programs Service rules on nodes.

THE SIGNAL

No node yet: the scheduler's one job.

> A developer writes a Service manifest but leaves out spec.type entirely.

What type of Service is created?

A NodePort

B LoadBalancer

C ExternalName

D ClusterIP

◆ PAUSE AND DECIDE

What type of Service is created?

A NodePort
Must be requested explicitly.

B LoadBalancer
Must be requested explicitly.

C ExternalName
Must be requested explicitly.

D ClusterIP
The default: a stable internal virtual IP.

THE SIGNAL

No type given: ClusterIP.

> A Pod runs an app container and a metrics-exporter sidecar.
>
> The exporter must read the app's statistics, which the app serves on port 9100.

How should the exporter address the app?

A localhost, on port 9100

B The name of a Service in front of the Pod

C The IP address of the node

D A separate IP belonging to the app container

◆ PAUSE AND DECIDE

How should the exporter address the app?

A localhost, on port 9100

Containers in a Pod share one network namespace.

B The name of a Service in front of the Pod

Works, but adds a needless hop through a Service.

C The IP address of the node

The node's IP does not reach the Pod's port.

D A separate IP belonging to the app container

Containers in a Pod have no separate IPs.

THE SIGNAL

Same Pod, shared network: localhost.

> Quayside runs Prometheus with its default design.

How does Prometheus normally collect metrics from a workload?

A The workload pushes its metrics to it over OTLP

B Fluent Bit forwards them from each node

C It scrapes each target's /metrics over HTTP

D The kubelet writes the metrics into etcd

◆ PAUSE AND DECIDE

How does Prometheus normally collect metrics from a workload?

A The workload pushes its metrics to it over OTLP
Push is possible, but not Prometheus's default model.

B Fluent Bit forwards them from each node
Fluent Bit ships logs, not metrics.

C It scrapes each target's /metrics over HTTP
Prometheus pulls on an interval.

D The kubelet writes the metrics into etcd
Metrics never go into etcd.

THE SIGNAL

Prometheus pulls.

> Quayside's stateless timetable service needs four identical replicas.
>
> New versions must roll out gradually, with the option to roll back.

Which workload fits?

A A StatefulSet

B A DaemonSet

C A Deployment

D A Job

◆ PAUSE AND DECIDE

Which workload fits?

A A StatefulSet
Adds identity and storage it does not need.

B A DaemonSet
One per node, not four identical copies.

C A Deployment
Identical replicas, rolling updates and rollback.

D A Job
Runs to completion; it would stop.

THE SIGNAL

Identical and stateless: Deployment.

> A new team member asks what a Helm chart actually is.

Which description is correct?

A. A package of templates and default values

B. One installed copy of an application

C. A repository that hosts packages

D. A numbered entry in an install's history

◆ PAUSE AND DECIDE

Which description is correct?

A. A package of templates and default values
The chart is the package.

B. One installed copy of an application
That is a release.

C. A repository that hosts packages
That is a chart repository.

D. A numbered entry in an install's history
That is a revision.

THE SIGNAL

Chart, release, repository, revision: four different nouns.

> An engineer wants to list only the Pods labelled tier=frontend, in the current namespace.

Which command does that?

A kubectl get pods --name=tier=frontend

B kubectl get pods -l tier=frontend

C kubectl describe pods tier=frontend

D kubectl get pods --field=tier=frontend

◆ PAUSE AND DECIDE

Which command does that?

A kubectl get pods --name=tier=frontend
No such flag for label filtering.

B kubectl get pods -l tier=frontend
-l takes a label selector.

C kubectl describe pods tier=frontend
Treats tier=frontend as a Pod name.

D kubectl get pods --field=tier=frontend
No such flag; field selectors work differently.

THE SIGNAL

Labels are selected with -l.

> A CI pipeline's ServiceAccount must create and update Deployments in the ci namespace, and nowhere else.

Which grant follows least privilege?

A A ClusterRole with a ClusterRoleBinding, for Deployments

B Adding the account to cluster-admin

C A Role and RoleBinding in default

D A Role and RoleBinding in the ci namespace

◆ PAUSE AND DECIDE

Which grant follows least privilege?

A A ClusterRole with a ClusterRoleBinding, for Deployments
Grants the power in every namespace.

B Adding the account to cluster-admin
Grants everything, everywhere.

C A Role and RoleBinding in default
Wrong namespace entirely.

D A Role and RoleBinding in the ci namespace
Exactly the verbs needed, in ci only.

THE SIGNAL

One namespace, specific verbs: Role plus RoleBinding there.

> On a kubeadm cluster, an engineer looks for the CoreDNS and kube-proxy Pods.

In which namespace will they be found?

A default

B kube-public

C kube-node-lease

D kube-system

◆ PAUSE AND DECIDE

In which namespace will they be found?

A default
For objects created without a namespace.

B kube-public
Readable by all; mostly unused.

C kube-node-lease
Holds node heartbeat Leases.

D kube-system
Objects created by Kubernetes itself.

THE SIGNAL

Kubernetes' own components live in kube-system.

> A single-replica internal database needs a persistent volume.
>
> It will be mounted read-write, by one node at a time.

Which access mode fits?

A ReadOnlyMany

B ReadWriteOnce

C ReadWriteMany

D ReadOnlyOnce

◆ PAUSE AND DECIDE

Which access mode fits?

A ReadOnlyMany

Read-only: the database must write.

B ReadWriteOnce

Read-write by one node at a time.

C ReadWriteMany

Many nodes: needs a shared filesystem it does not need.

D ReadOnlyOnce

Not a real access mode.

THE SIGNAL

One node, read-write: ReadWriteOnce.

> The booking API should get more Pods automatically when average CPU rises, and fewer when it falls.

Which object does this?

A A VerticalPodAutoscaler

B A PodDisruptionBudget

C A HorizontalPodAutoscaler

D A ResourceQuota

◆ PAUSE AND DECIDE

Which object does this?

A A VerticalPodAutoscaler
Changes Pod size, not count.

B A PodDisruptionBudget
Limits voluntary disruptions.

C A HorizontalPodAutoscaler
Adjusts the replica count from metrics.

D A ResourceQuota
Caps a namespace's totals.

THE SIGNAL

More or fewer Pods from metrics: HPA.

> New GPU nodes must repel every Pod that does not explicitly tolerate them.
>
> Pods already running on those nodes may stay.

What should be applied to the GPU nodes?

A A taint with the NoSchedule effect

B A taint with the NoExecute effect

C A nodeSelector on the GPU nodes

D A high PriorityClass

◆ PAUSE AND DECIDE

What should be applied to the GPU nodes?

A A taint with the NoSchedule effect
Repels new Pods; leaves running ones.

B A taint with the NoExecute effect
Would also evict running Pods.

C A nodeSelector on the GPU nodes
Selectors go on Pods, and attract rather than repel.

D A high PriorityClass
Controls order and preemption, not placement.

THE SIGNAL

Repel new Pods, keep running ones: NoSchedule.

> A platform lead explains GitOps to a new team.

How do changes reach the cluster in GitOps?

A An agent in the cluster pulls from Git and reconciles

B The CI pipeline runs kubectl apply after every build

C Engineers edit live objects, then commit to Git

D The scheduler reads manifests straight from Git

◆ PAUSE AND DECIDE

How do changes reach the cluster in GitOps?

A An agent in the cluster pulls from Git and reconciles

Pulled automatically, continuously reconciled.

B The CI pipeline runs kubectl apply after every build

That is push-based delivery.

C Engineers edit live objects, then commit to Git

Backwards: Git must lead, not follow.

D The scheduler reads manifests straight from Git

The scheduler only places Pods.

THE SIGNAL

GitOps pulls.

> A brand new namespace has no NetworkPolicies at all.

Which traffic can reach its Pods?

A None: namespaces deny by default

B Only traffic from the same namespace

C Traffic from any Pod in the cluster

D Only traffic from kube-system

◆ PAUSE AND DECIDE

Which traffic can reach its Pods?

A None: namespaces deny by default

Default deny must be created as a policy.

B Only traffic from the same namespace

Namespaces are not network boundaries by default.

C Traffic from any Pod in the cluster

Pods are non-isolated until a policy selects them.

D Only traffic from kube-system

Nothing special applies to kube-system.

THE SIGNAL

No policy selects a Pod: everything allowed.

> A student compares containers with virtual machines.

Which statement is correct?

A Each container runs its own kernel

B Containers need a hypervisor to run

C Containers boot a full guest operating system

D Containers share the host's kernel

◆ PAUSE AND DECIDE

Which statement is correct?

A Each container runs its own kernel
That describes virtual machines.

B Containers need a hypervisor to run
That describes virtual machines.

C Containers boot a full guest operating system
That describes virtual machines.

D Containers share the host's kernel
The key difference from VMs.

THE SIGNAL

Shared kernel: the whole difference.

> A booking API Pod's readiness probe starts failing. Its liveness probe keeps passing.

What happens to the Pod?

A The container is restarted

B It keeps running, but gets no Service traffic

C Its node is drained

D The Deployment rolls back to its previous revision

◆ PAUSE AND DECIDE

What happens to the Pod?

A The container is restarted

Only a failing liveness probe restarts.

B It keeps running, but gets no Service traffic

Readiness removes it from endpoints.

C Its node is drained

Probes never drain nodes.

D The Deployment rolls back to its previous revision

No automatic rollback happens.

THE SIGNAL

Readiness removes from traffic; liveness restarts.

> The web front end needs a non-confidential setting, a timetable refresh interval of 300 seconds, kept outside the image.

Where should the setting live?

A A Secret

B An annotation on the Deployment

C A ConfigMap

D A PersistentVolumeClaim

◆ PAUSE AND DECIDE

Where should the setting live?

A A Secret

For confidential data.

B An annotation on the Deployment

Describes the object; not consumed as config.

C A ConfigMap

Non-confidential configuration.

D A PersistentVolumeClaim

Storage for data, not settings.

THE SIGNAL

Non-confidential config: ConfigMap.

> Pods resolve names such as web.quayside.svc.cluster.local to Service addresses.

Which component answers those lookups?

A kube-proxy

B The kubelet

C etcd

D CoreDNS

◆ PAUSE AND DECIDE

Which component answers those lookups?

A kube-proxy

Programs traffic rules, not DNS.

B The kubelet

Points Pods at DNS, but does not answer lookups.

C etcd

Stores state; answers no DNS.

D CoreDNS

The cluster DNS server.

THE SIGNAL

Cluster DNS: CoreDNS.

> A team wants one vendor-neutral way to instrument its services for traces, metrics and logs.

Which CNCF project provides that?

A OpenTelemetry

B Jaeger

C Prometheus

D Fluentd

◆ PAUSE AND DECIDE

Which CNCF project provides that?

A **OpenTelemetry**
APIs, SDKs and OTLP for all three signals.

B **Jaeger**
A tracing backend.

C **Prometheus**
Metrics collection and storage.

D **Fluentd**
Log collection and routing.

THE SIGNAL

All three signals, vendor-neutral: OpenTelemetry.

> A monthly data-cleanup script must run once to completion, and be retried if it fails.

Which workload fits?

A A Deployment

B A Job

C A DaemonSet

D A StatefulSet

◆ PAUSE AND DECIDE

Which workload fits?

A A Deployment
Restarts the script after it succeeds.

B A Job
Runs to completion, with retries.

C A DaemonSet
One copy per node, running forever.

D A StatefulSet
For stateful long-running services.

THE SIGNAL

Finite work: Job.

Score yourself by domain

DOMAIN	QUESTIONS	YOUR SCORE
Kubernetes Fundamentals (44%)	1, 3, 5, 7, 9, 12, 15, 17, 20	of 9
Container Orchestration (28%)	2, 8, 10, 14, 18	of 5
Application Delivery (16%)	6, 13, 16	of 3
Cloud Native Architecture (12%)	4, 11, 19	of 3

ON THE EXAM

Which domain your misses cluster in matters more than your total.

What your score means

SCORE	MEANS	NEXT STEP
17 to 20 (85%+)	Ready, with margin	Keep going: practice exam 2
15 or 16 (75 to 80%)	On the pass line	Rewatch the modules behind your misses
Below 15	Not yet	Revisit the weakest domain's modules, then retake

ON THE EXAM

The real pass mark is 75%. Aim for a steady 85% in practice, to leave margin for nerves.

Practice exam 2: twenty new questions, no topic repeated from this paper.

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